

# Low-Light Image Enhancement with Deep Learning

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# Original Proposal Scope

## Initial Plan:

- Compare CNN architectures for low-light enhancement
- Evaluate using:
  - PSNR (Peak Signal-to-Noise Ratio)
  - SSIM (Structural Similarity)
- Use RGB images only
- Train multiple architectures

This project initially focused only on RGB enhancement and image quality metrics.

# Updated Project Scope

## Final Expanded Scope:

- RGB + Grayscale input comparisons
- Multiple architectures:
  - Shallow CNN
  - Deep CNN
  - U-Net
  - ResUNet
- Added:
  - Binary classification task (downstream evaluation)
- Full pipeline:
  - Enhancement → Classification

This expansion allows evaluation of both image quality and downstream task performance.

# Introduction to the problem

## Current Challenges

- Low-light images often contain weak illumination, low contrast, color distortion, and heavy sensor noise
- Important visual details such as edges, textures, and object boundaries are often degraded or lost
- These degradations make enhancement difficult and also reduce downstream recognition and classification performance

## Problem Statement

- We want to enhance low-light images in a way that improves both image quality and downstream binary classification
- A visually brighter or sharper image is not always the most useful image for semantic recognition
- The key question is which enhancement architecture best balances restoration quality and practical task performance

# Project Pipeline

## Pipeline Overview:

1. Input: Low-light images
2. Enhancement model
3. Output: Enhanced images
4. Binary classifier evaluates usefulness

## Key Idea:

Better enhancement → better classification performance

- Binary classifier trained on enhanced outputs
- Used to evaluate practical usefulness

# Models Used

## Enhancement Models:

- Shallow CNN
- Deep CNN
- U-Net
- ResUNet

## Input Modes:

- RGB
- Grayscale

Shallow CNN → baseline model

Deep CNN → increased capacity

U-Net → encoder-decoder with skip connections

ResUNet → residual + skip connections for deeper learning

# Evaluation Metrics

## Image Quality Metrics:

- PSNR  $\uparrow$  (higher is better)
- SSIM  $\uparrow$  (closer to 1 is better)

## Training Metrics:

- Loss  $\downarrow$

## Downstream Task:

- Binary classification accuracy

- PSNR measures reconstruction quality (pixel-level fidelity)
- SSIM measures perceptual similarity (structure preservation)

# Experimental Refinement

## Problem:

- Initial runs used mixed training configurations
- Some models required stabilization (NaN issues)
- Observed differences in training behavior across models
- Re-trained all models using consistent, stable settings

## Causes:

- Mixed precision (AMP)
- High learning rate
- Complex loss (L1 + Charbonnier + SSIM)

**To ensure fair comparison, all models were retrained under identical stable conditions**

# Fixes Applied

## Stabilization Strategy:

- Learning rate:  $1e-5$
- Loss: L1 only
- AMP: disabled for unstable models
- Gradient clipping: enabled
- Epochs: 20
- Batch size: 4

## Result:

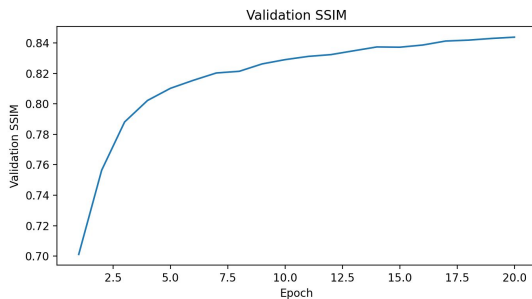
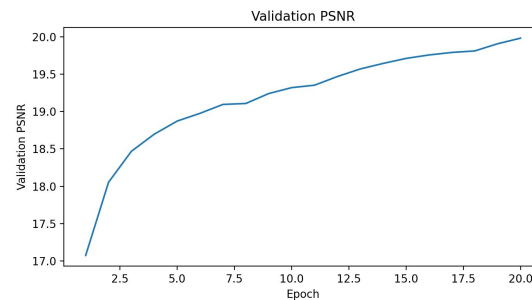
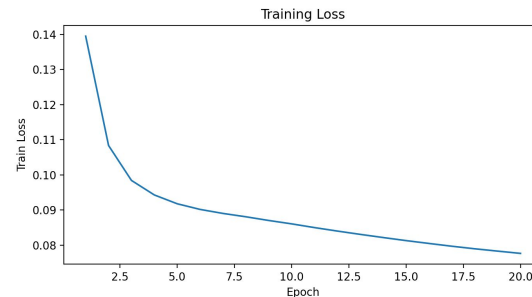
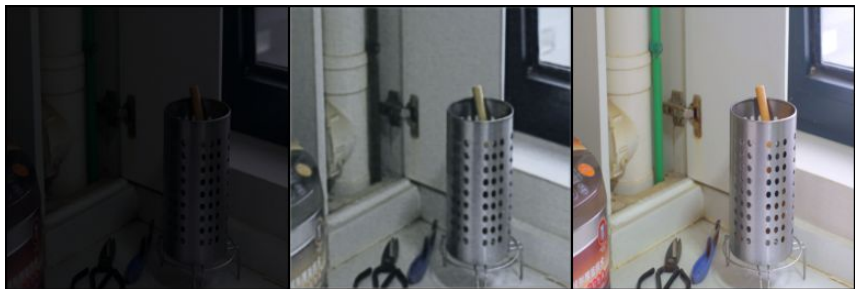
- Stable training
- No NaN values

These changes reduced gradient explosion and improved numerical stability.

# Shallow CNN (RGB)

- PSNR:  $\sim 19.98$
- SSIM:  $\sim 0.84$

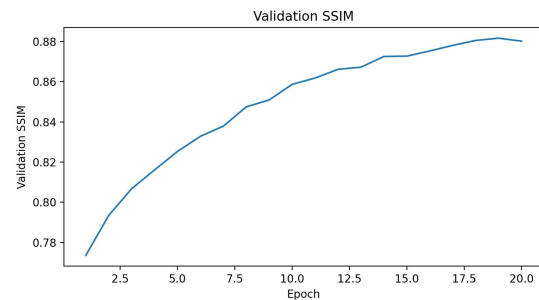
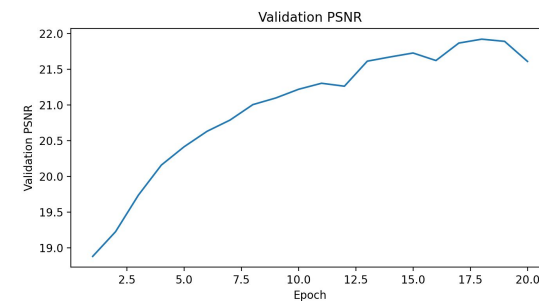
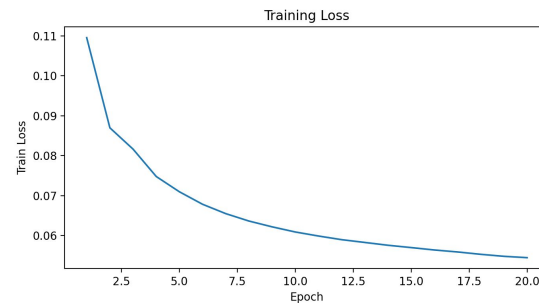
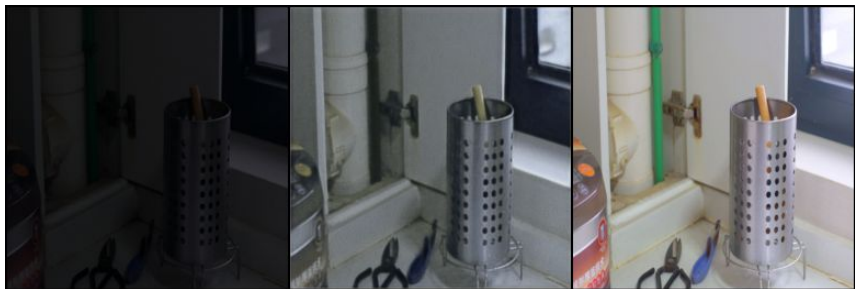
Limited performance due to shallow architecture and lack of feature hierarchy.



# Deep CNN (RGB)

- PSNR:  $\sim 21.62$
- SSIM:  $\sim 0.88$

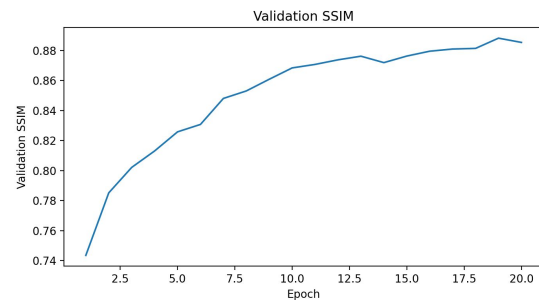
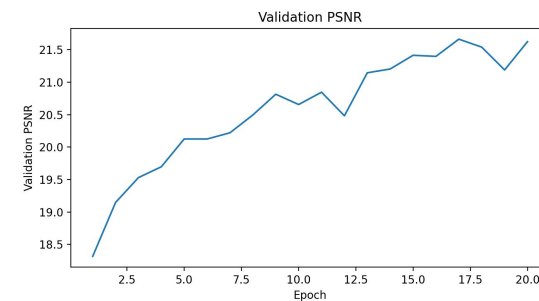
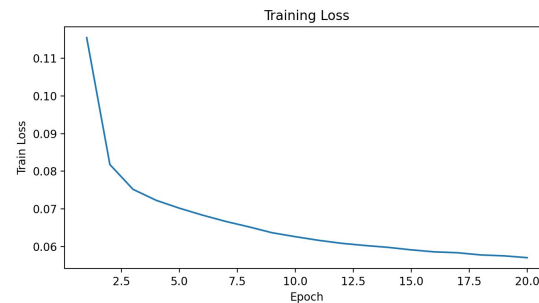
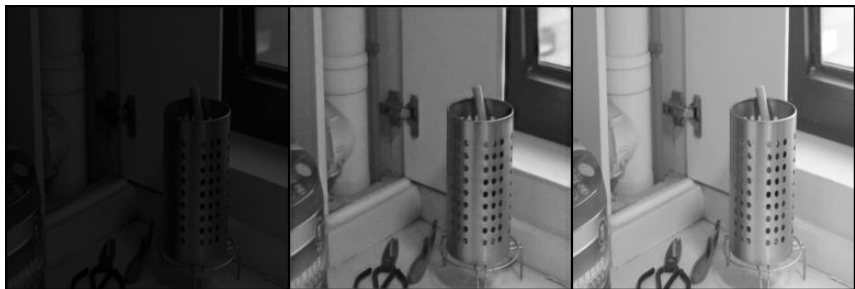
Improved over shallow CNN due to deeper feature extraction.



# Deep CNN (Gray)

- PSNR:  $\sim 21.62$
- SSIM:  $\sim 0.89$

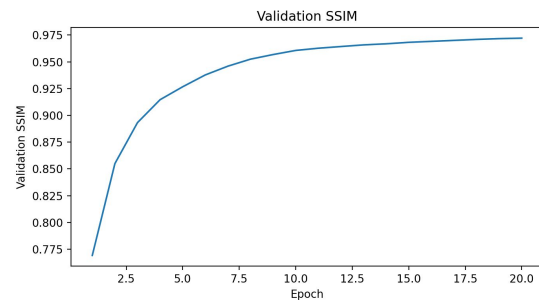
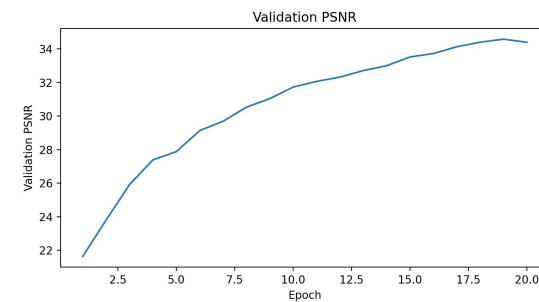
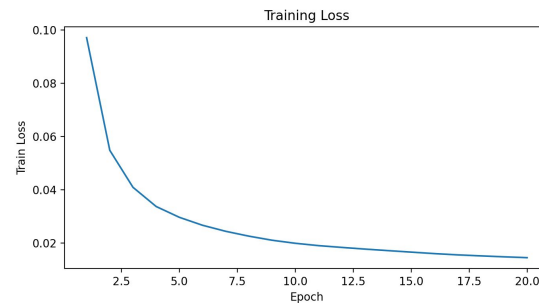
Grayscale removes color information, slightly limiting performance.



# U-Net (RGB)

- PSNR:  $\sim 34.40$
- SSIM:  $\sim 0.97$

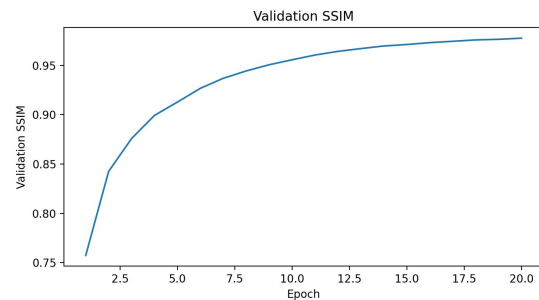
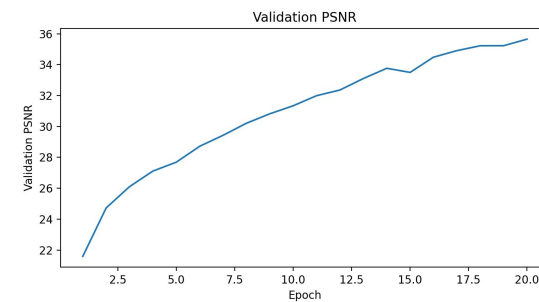
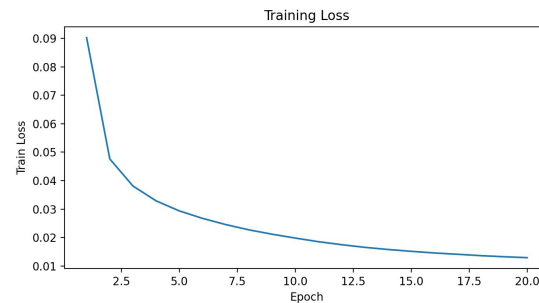
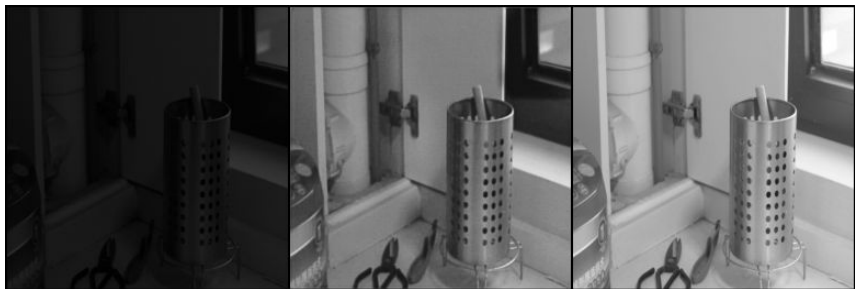
**Significant improvement due to skip connections preserving spatial details.**



# U-Net (Gray)

- PSNR:  $\sim 35.65$
- SSIM:  $\sim 0.98$

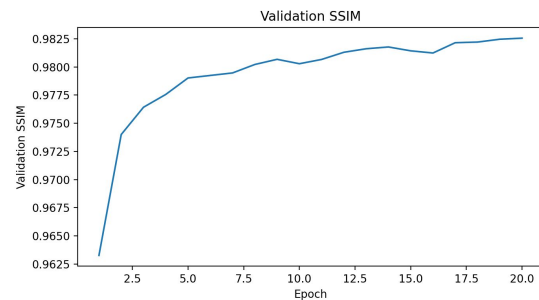
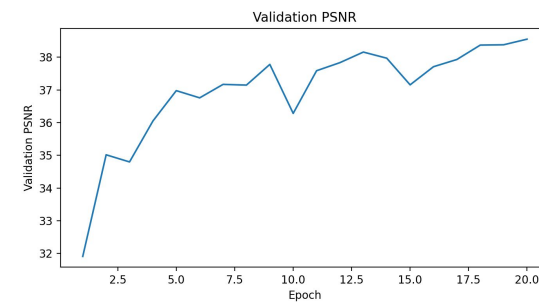
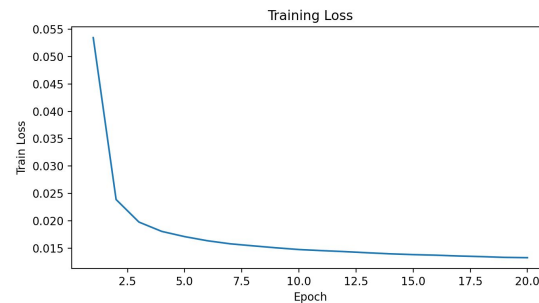
**Performs slightly better than RGB due to reduced complexity and noise.**



# ResUNet (RGB — Retry)

- PSNR:  $\sim 38.55$
- SSIM:  $\sim 0.98$

Best performance due to residual learning and deep architecture.



# Comparison Table

| Model       | PSNR  | SSIM |
|-------------|-------|------|
| Shallow RGB | 19.98 | 0.84 |
| Deep RGB    | 21.62 | 0.88 |
| Deep Gray   | 21.62 | 0.88 |
| U-Net RGB   | 34.40 | 0.97 |
| U-Net Gray  | 35.65 | 0.98 |
| ResUNet RGB | 38.55 | 0.98 |

ResUNet achieves the highest PSNR and SSIM, indicating superior reconstruction quality.

# Why Models Performed Differently

## **Better Models (U-Net, ResUNet):**

- Skip connections → preserve detail
- Deeper architecture → better feature learning

## **Worse Models (Shallow):**

- Limited capacity
- Poor generalization

## **Gray vs RGB:**

- RGB retains more information

This demonstrates the importance of architectural design in low-light enhancement tasks.

# Downstream Binary Classification

- Task: classify images into two categories
- Classes:
  - Class 0: Low-quality / low-light images
  - Class 1: High-quality / well-lit images
- Purpose:
  - Evaluate whether enhanced images are more useful for downstream tasks
  - Measure if enhancement improves semantic understanding

Classifier is trained on enhanced outputs from each model. Higher accuracy indicates better enhancement quality in a practical setting.

# Downstream Binary Classification Results

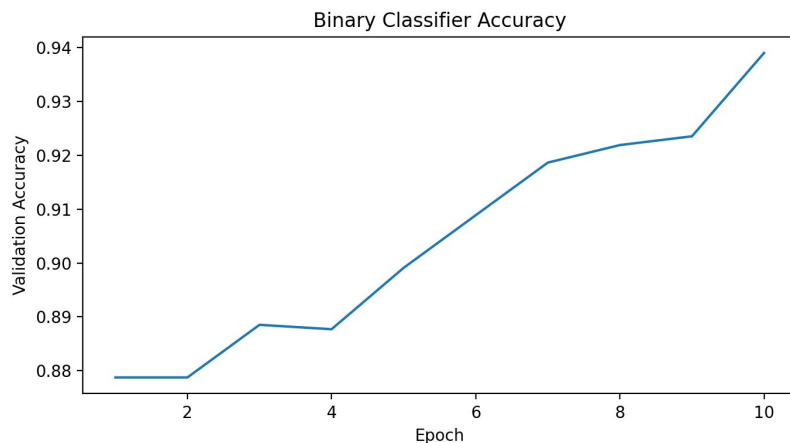
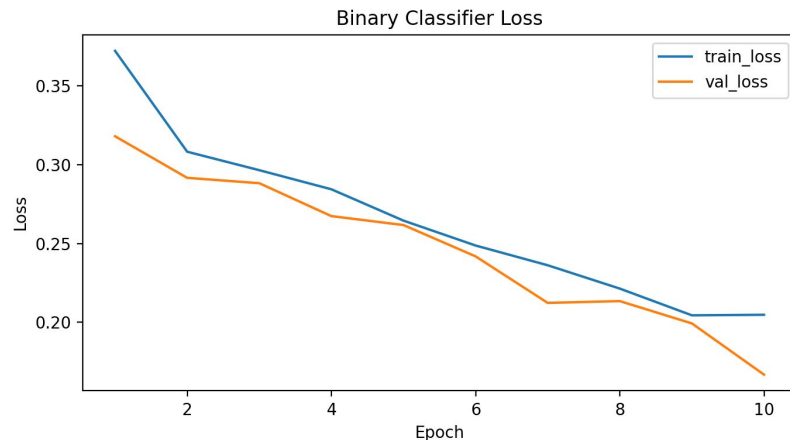
## Purpose:

Evaluate usefulness of enhanced images

## Observation:

- Enhanced images improve classification accuracy
- Best improvements from:
  - ResUNet RGB
  - U-Net RGB

- Classifier accuracy:  $\sim 0.94$
- Shows enhancement improves downstream ML tasks



# Key Insights

- Deep models outperform shallow models
- Skip connections are critical
- RGB inputs perform better than grayscale
- Stable training is essential for complex architectures
- Standardizing training conditions was critical for fair comparison
- Enhancement improves downstream classification

# Future Work

- Add perceptual loss
- Use GAN-based enhancement
- Larger datasets
- Real-time deployment

Future work will focus on perceptual quality and real-world deployment.

# Conclusion

- Successfully built full enhancement pipeline
- Fixed major training instability issues
- Demonstrated improvement in both:
  - Image quality
  - Classification performance

ResUNet and U-Net are most effective for low-light enhancement tasks.

# Thank You

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